

Jayani Sri Nandika Duggirala

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science in Computer Science

2025-2029

Coursework: Computer Vision, Data Structures & Algorithms, Multi-variable Calculus, Linear Algebra, Probability & Statistics, Object-Oriented Programming

EXPERIENCE

CSE Society (CSES) @ UCSD | *Researcher & Developer – ORCA*

Oct. 2025 - April 2026

- Individually researched and summarized papers on self-supervised music representation learning (including S-KEY), automatic music transcription architectures (Onsets and Frames, MR-MT3), and generative source separation (GuideSep), among others, to inform team research direction.
- Co-authored a literature review evaluating SOTA AMT architectures across polyphonic accuracy, multi-instrument generalization, and computational cost tradeoffs
- Proposed AMT pipeline: audio source separation → per-instrument MR-MT3 embedding extraction → music-theory-constrained decoding → MIDI/MusicXML output, then applying domain music-theory knowledge to constrain generation rather than the existing LLM-only post-processing approaches to improve deep learning architectures for ORCA (Orchestration and Recognition for Composition and Arrangement), an AI-assisted music recognition and creation tool.

ACM AI @ UCSD

May 2026 - Present

AI Events Lead

- Designing technical workshop curriculum (AI School series) covering classification, logistic regression, neural network fundamentals, and evaluation, taught via Google Colab notebooks and live presentations to club members
- Built hands-on teaching materials using scikit-learn and Keras across supervised learning workshop modules
- Hosted community hours for Women in Computing (WIC) @ UCSD, teaching development environment setup (virtual environments, tooling basics), and the python library Manim (CE) to new programmers.

PUBLICATIONS

- Duggirala, J., Hiremath, S., & Goyal, U. (2025). *Automatic Music Transcription (AMT) – Literature Review*. CSES ORCA. [PDF]

PROJECTS

Audiate | Python, CREPE, librosa, Swift

April 2026

- Built a system that transcribes hummed/sung melodies or full symphonic audio into an animated MIDI representation, mapping vocal input to the nearest semitone and visualizing bow-stroke direction from rhythm data, submitted at SDx UCSD Hackathon
- Used **CREPE** (deep learning pitch detection) for pitch estimation and **librosa** for rhythm analysis; built Swift-based haptic feedback synced to the note timeline to improve music accessibility for deaf and hard-of-hearing users. [GitHub](#)

AWARDS/COMPETITIONS

- Participant, 2nd Annual NSF HDR & UCSD SMASH Hackathon : NSF HDR ML Challenge, Climate Prediction (Ecological Data) track
- 1st Place, DataHacks 2026 @ UCSD (400+ participants) : Economics track
- 1st Place, UCSD Programming Competition for Underclassmen hosted by WIC, Winter 2026
- 3rd Place, SDx UCSD Hackathon
- CIF Division 1 SJS Section Champion, 2022 & 2024 (top among 206 HS): #1 Singles Tennis
- 1st Place, WIC Fall Project Teams 2025 : Built Matchstick, a full-stack peer study-matching platform

TECHNICAL SKILLS

Languages: Python, Java, Swift, C

ML/DL: TensorFlow, PyTorch, scikit-learn, Keras, pandas, NumPy, matplotlib, Hugging Face

Developer Tools: GitHub, Jupyter Notebook, Google Colab, Kaggle